

BINF 5503

Data Analytics and Storytelling

Week 5

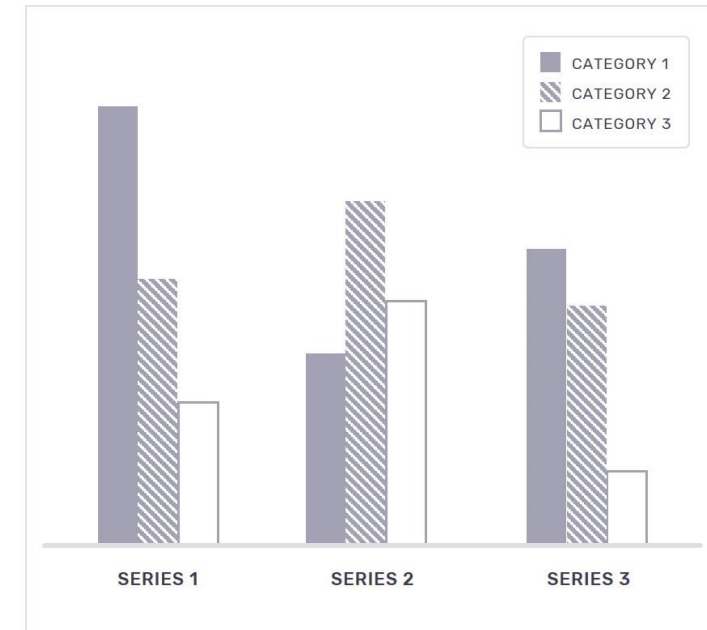
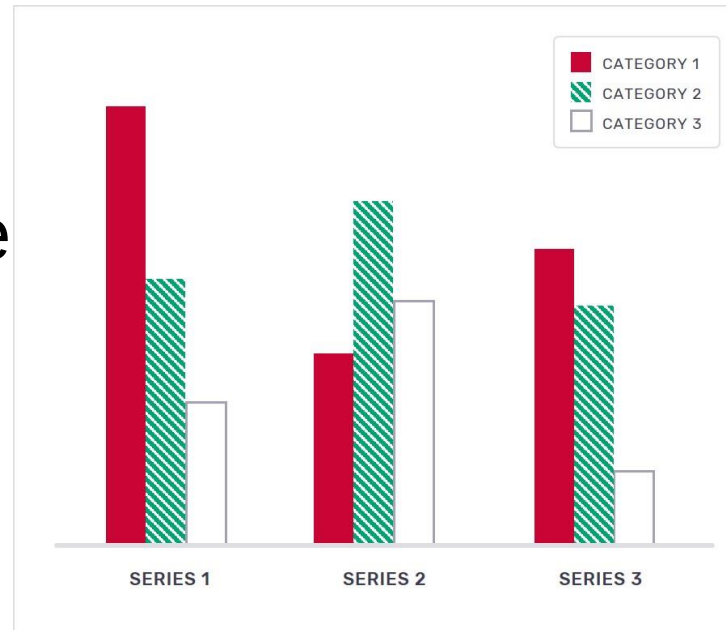
Module 4 – Giving Presentations, Writing and Critiquing Reports

Outline

- Presentations
 - Accessibility
 - Other tips
- Writing and presenting reports
 - Introduction
 - Methods & Results
 - Discussion & Conclusions
- Critiquing reports and presentations
 - Identify common errors
 - Evaluate real posters

Making your poster accessible to a diverse audience – visual considerations

- Ensure enough luminance contrast between colours
 - E.g., black/dark blue on white background; or, white text on black/dark blue background
 - Contrast grid: <https://contrast-grid.eightshapes.com/>
- Use non-serif fonts at a large font size
- Do not use colour as the only way to define information



Making your poster accessible to a diverse audience – visual considerations

- Avoid colour combinations that are common colour-blindness combinations
 - Red/green, blue/yellow
 - Online simulator: <https://www.color-blindness.com/coblis-color-blindness-simulator/>



REAL COLORS



RED-GREEN
BLINDNESS



BLUE-YELLOW
BLINDNESS



MONOCHROMACY
BLINDNESS

Making your poster accessible to a diverse audience – oral/aural considerations

- Avoid using too many idioms, acronyms, or jargon
- Indicate if there will be closed captioning, transcripts, or sign language
- Order of visual presentation == order of oral presentation
- Speak louder and slower than you think!



Making your poster accessible to a diverse audience – tables and figures

- All graphics have descriptive captions that can be standalone
- Select the best medium for different types of information: text, tables, or figures
 - Information that does not add to your argument does not need to be included
- If you are adding text *over* images, use a solid overlay behind the text

Making your poster accessible to a diverse audience – other tips



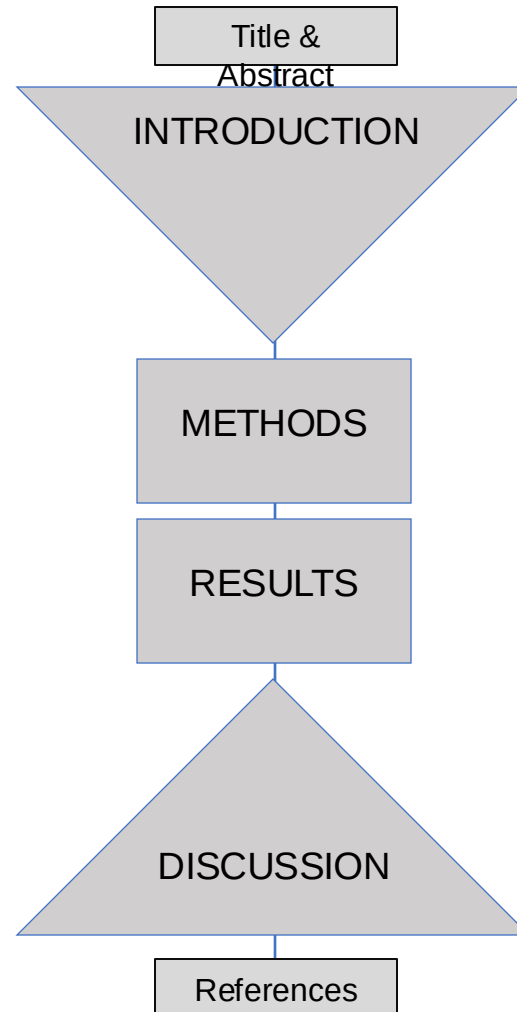
- There is some kind of flow
- Use icons as visual cues if it connects to concepts
- Key information is clear at a glance
- Consider separating text into columns

Next: reading & writing reports

- Become familiar with the structure and content of scientific writing
- Become comfortable reading and identifying important scientific information
- Think critically about information presented in science
- Apply knowledge of scientific writing structure to your own report and final presentation

Structure of a scientific report

- Title
- Abstract
- Introduction
- Methods
- Results
- Discussion
- References



Structure of a scientific report

- **Title**

- Descriptive; indicates the purpose of the paper.

- **Abstract**

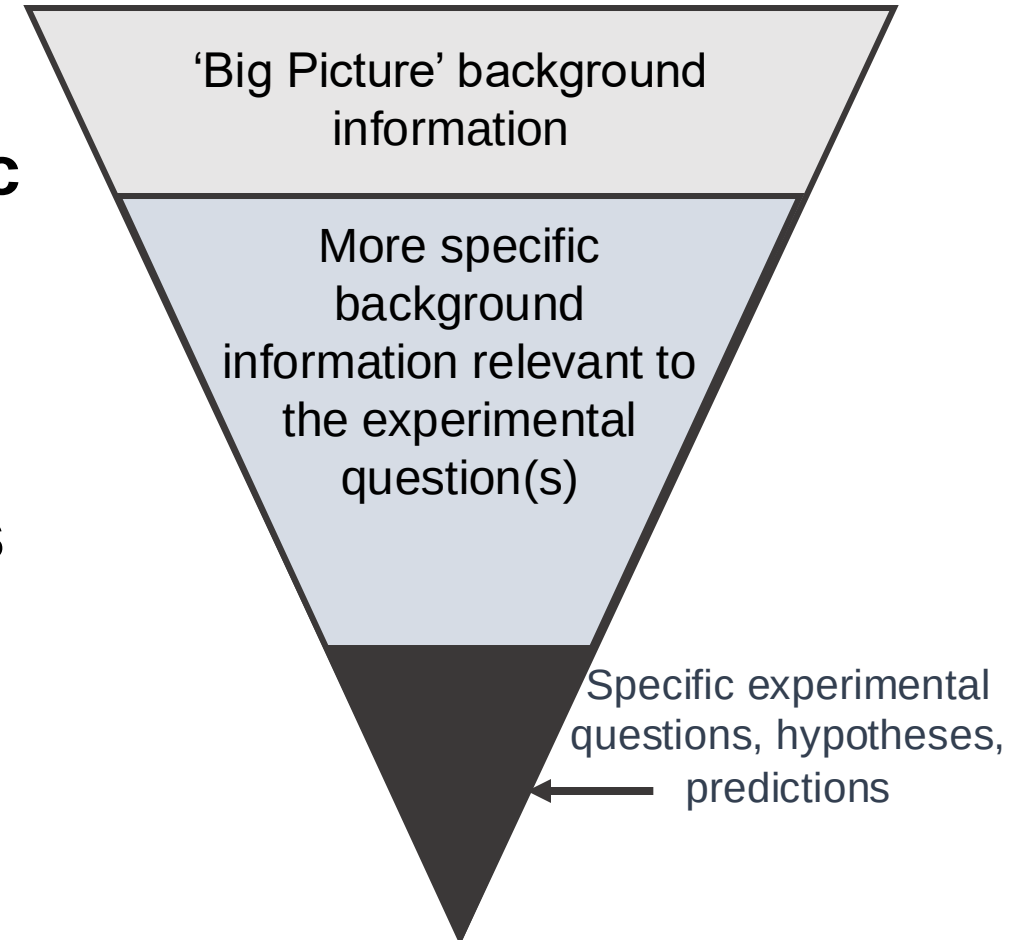
- Brief, comprehensive summary of the purpose, main results and conclusion(s).

- **References**

- Provides the sources cited throughout the article.

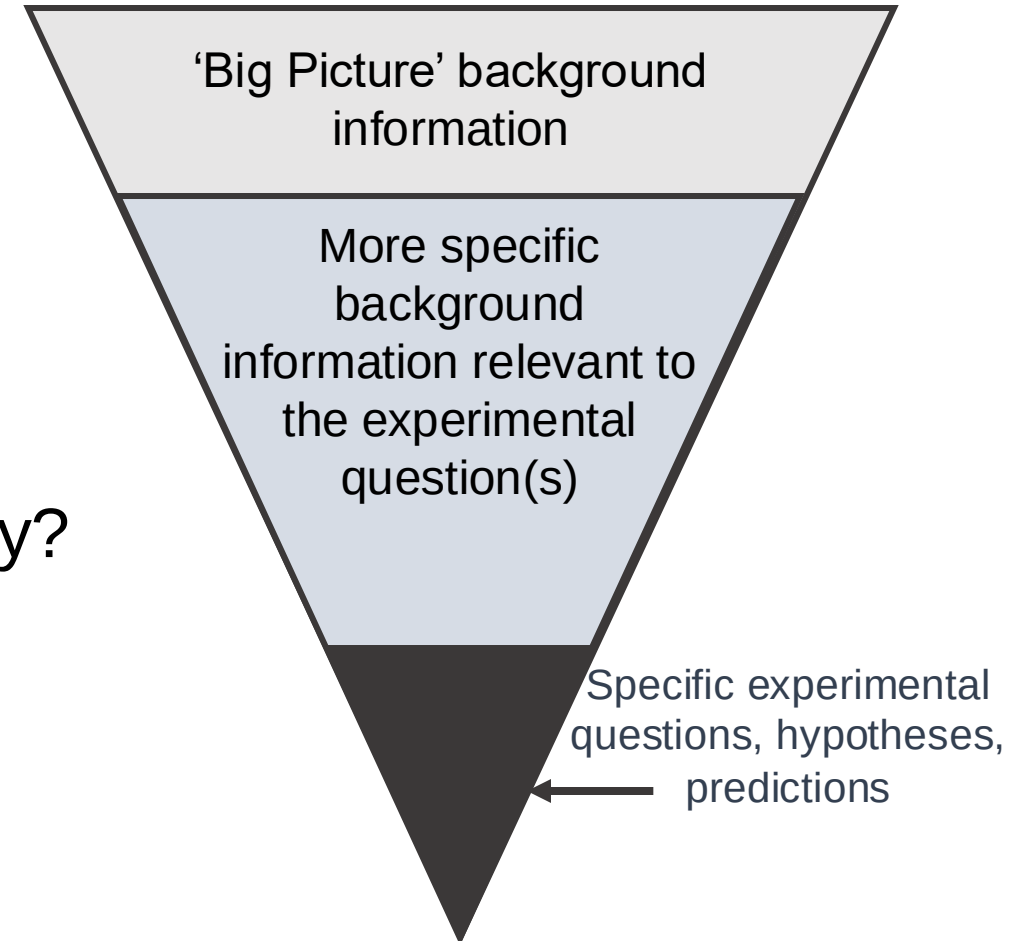
Introduction

- Provides the **framework and scientific context** for experimental question(s), hypotheses and predictions
 - Justifies the study
- Generally includes a comprehensive **literature review of relevant previous studies** (with appropriate 'in-text' citations)
- Overall structure is from **broad**, “Big Picture” background concepts and information to **specific experimental questions, hypotheses and predictions**



Introduction

- What main, “big picture” concepts are presented in the introduction?
- What other main points/ideas are presented in the introduction?
- What is the purpose of the current study?
- What are the specific hypothesis statements and/or predictions?



Methods

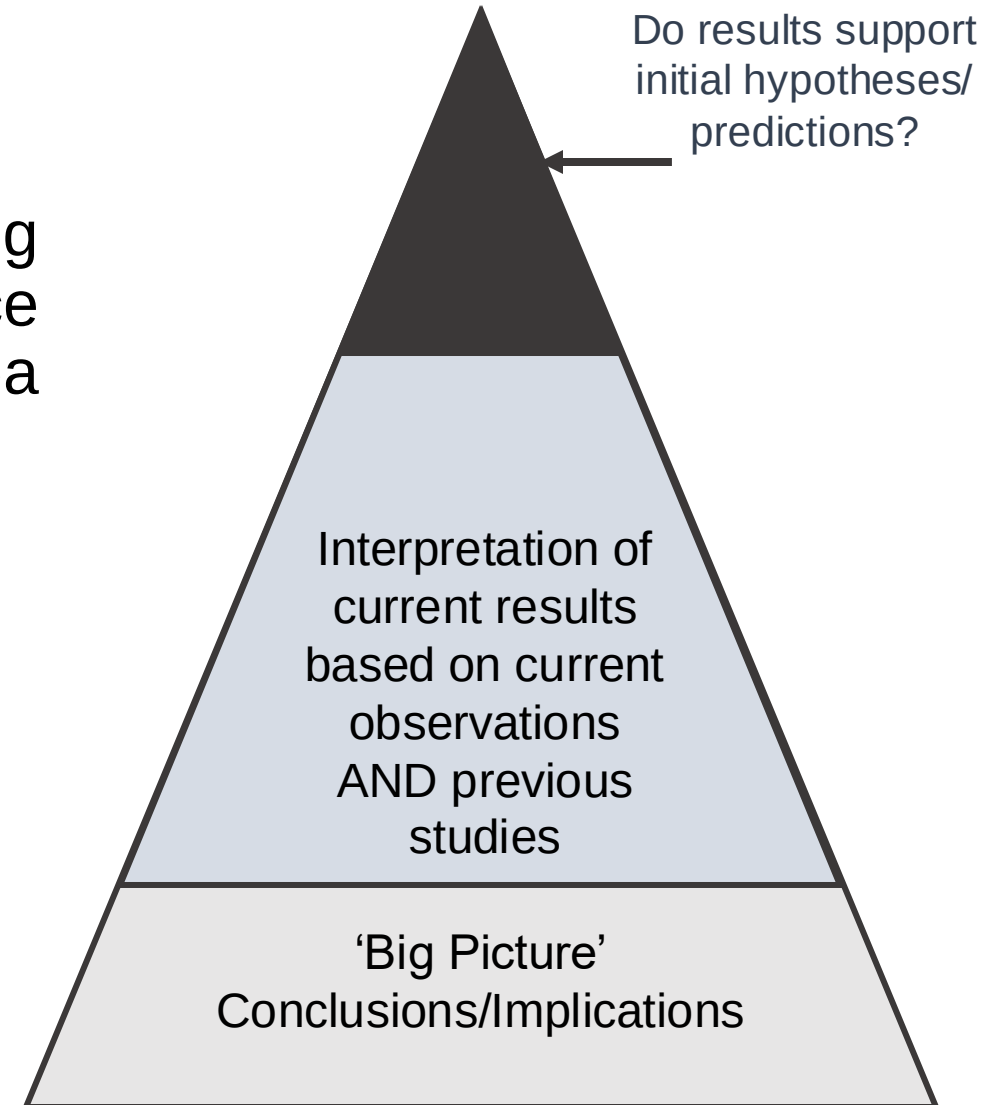
- What is the purpose of the current study?
- Where was the study area? What type of data was collected?
- What comparisons were made? How was the data analyzed?

Results

- The most important section of a scientific writing
- **Clearly** and **concisely** presents the results of the experiment(s) conducted; generally with a combination text, tables and/or figures
- State trends but do not discuss here

Discussion

- Results are interpreted for scientific meaning and their significance/ importance/ relevance to the existing body of knowledge in the area of study
 - Were the hypotheses supported?
- Generally refers extensively to **relevant previous studies** to help explain and support current results
- Overall structure is from **specific** statement(s) about whether the results supported the initial hypotheses/predictions to **broad** 'Big Picture' conclusions and scientific implications



In scientific writing, avoid:

- Filler words & phrases:
 - “nuanced”, “intricate”, “absolutely”, “definitely”, etc.
- Passive voice
 - “a linear regression was performed” versus “we used a linear regression to model...”
- Direct quotes
- Inconsistent past/present tense

Next: critiquing reports

- Look at some writing examples and poster examples
- Evaluate how the information is being presented
- Justify good and poor practices from reports
- Apply knowledge of scientific writing to improve examples provided

Let's look at some scientific writing examples

Examples taken from the Robert Gillespie Academic Skills Centre (RGASC) at UTM.

- Some examples have grammatical errors
- Some examples have issues that are harder to find
- Make improvements

What's wrong with these examples?

“Significant improvements in light environments were observed in artificially created gaps in the forest canopy. Thus resulting in better growth performance of the light-demanding plants on the forest floor.”

“In a study looking at two canopy gap sizes in central New Zealand, Forbes et al. (2016) found significantly higher natural regeneration of light-demanding plants than planted seedlings in gaps with at least 50% light transmission. Although larger canopy gaps with 84% light transmission were more favorable for planted seedlings.”

Sentence fragment

“Significant improvements in light environments were observed in artificially created gaps in the forest canopy. Thus resulting in better growth performance of the light-demanding plants on the forest floor.”

“In a study looking at two canopy gap sizes in central New Zealand, Forbes et al. (2016) found significantly higher natural regeneration of light-demanding plants than planted seedlings in gaps with at least 50% light transmission. Although larger canopy gaps with 84% light transmission were more favorable for planted seedlings.”

Sentence fragment - fixes

“Significant improvements in light environments were observed in artificially created gaps in the forest canopy. This resulted in better growth performance of the light-demanding plants on the forest floor.”

“In a study looking at two canopy gap sizes in central New Zealand, Forbes et al. (2016) found significantly higher natural regeneration of light-demanding plants than planted seedlings in gaps with at least 50% light transmission. However, larger canopy gaps with 84% light transmission were more favorable for planted seedlings.”

What's wrong with this example?

“At the forest edge, tree growth was highly variable, some seedlings grew as tall as or taller than trees in the canopy gap.”

Comma splice

“At the forest edge, tree growth was highly variable, some seedlings grew as tall as or taller than trees in the canopy gap.”

Comma splice

“At the forest edge, tree growth was highly variable, some seedlings grew as tall as or taller than trees in the canopy gap.”

At the forest edge, tree growth was highly variable, since some seedlings grew as tall as or taller than trees in the canopy gap.

At the forest edge, tree growth was highly variable; some seedlings grew as tall as or taller than trees in the canopy gap.

At the forest edge, tree growth was highly variable. Some seedlings grew as tall as or taller than trees in the canopy gap.

What's wrong with this example?

“One example of competitive exclusions of early successional species in regenerating forests is where low-light environments prevent the establishment of late-successional species, which, in some environments, can persist for many decades before changes in canopy structure allow light environments to increase and recruitment to occur, which gives rise to the need for gap-based interventions to create a more open stand structure in the restoration of many shade-intolerant late-successional species, including oak and pine, although there is concern about the intermittent effect of gaps on the restoration of shade intolerant species as the process of gap closure could reduce plant growth over time and restrict seedling development into the canopy.”

Run-on sentence

“One example of competitive exclusions of early successional species in regenerating forests is where low-light environments prevent the establishment of late-successional species, which, in some environments, can persist for many decades before changes in canopy structure allow light environments to increase and recruitment to occur, which gives rise to the need for gap-based interventions to create a more open stand structure in the restoration of many shade-intolerant late-successional species, including oak and pine, although there is concern about the intermittent effect of gaps on the restoration of shade intolerant species as the process of gap closure could reduce plant growth over time and restrict seedling development into the canopy.”

Run-on sentence

“One example of competitive exclusions of early successional species in regenerating forests is where low-light environments prevent the establishment of late-successional species. In some cases, these lowlight environments persist for many decades before changes in canopy structure allow light environments to increase and recruitment to occur. This gives rise to the need for gap-based interventions to create a more open stand structure in the restoration of many shade intolerant late-successional species, including oak and pine. However, there is concern about the intermittent effect of gaps on the restoration of shade-intolerant species; the process of gap closure could reduce...”

What's wrong with these examples?

“The cluster of trees that survived the low-light environment long-term were more tolerant of shade than tōtara.”

“Estimation of percent light transmission from the hemispherical photos were accomplished using Gap Light Analyzer.”

Subject-Verb Agreement

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“Estimation of percent light transmission from the hemispherical photos was accomplished using Gap Light Analyzer.”

What's wrong with these examples?

“The results of this study suggested that gap creation was a productive tool for restoring late successional canopy species in regenerating forests.”

“The trial is established in October 2009 in a regenerating kanuka forest.”

“Restoration of degraded lands was typically undertaken by facilitating natural regeneration.”

Improper tense use

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Improper tense use

“The results of this study suggest that gap creation was a productive tool for restoring late successional canopy species in regenerating forests.”

“The trial was established in October 2009 in a regenerating kanuka forest.”

“Restoration of degraded lands has typically been undertaken by facilitating natural regeneration.”

What's wrong with these examples?

“Each replicate comprised 20 locally sourced, nursery-raised tōtara seedlings planted approximately 1m apart, in four rows of five seedlings. They were located within an altitudinal band of 100–200m a.s.l. but where possible keeping slope and aspect similar.”

“A common means of restoring abandoned farmland to forest is to facilitate natural regeneration. To do this, they remove degrading factors from the site, such as grazing animals.”

Unclear pronoun antecedent: “they”

“Each replicate comprised 20 locally sourced, nursery-raised tōtara seedlings planted approximately 1m apart, in four rows of five seedlings. They were located within an altitudinal band of 100–200m a.s.l. but where possible keeping slope and aspect similar.”

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“Each replicate comprised 20 locally sourced, nursery-raised tōtara seedlings planted approximately 1m apart, in four rows of five seedlings. All replicates were located within an altitudinal band of 100–200m a.s.l. but where possible keeping slope and aspect similar.”

“A common means of restoring abandoned farmland to forest is to facilitate natural regeneration. To do this, restoration managers remove degrading factors from the site, such as grazing animals.”

Or: “... To do this, the site’s degrading factors, such as grazing animals, must be removed.”

What's wrong with these examples?

“The greatest variability in growth and lower seedling survival was found at the forest edge sites.”

“The highest light transmission was provided by artificial gaps from selective felling of kanuka.”

“In one study, it was reported by Ramos and del Amo (1992) that consistent growth improvement and highest survival of all planted seedlings in a canopy treatment with 37% light transmission.”

Overuse of the passive voice

“The greatest variability in growth and lower seedling survival was found at the forest edge sites.”

“The highest light transmission was provided by artificial gaps from selective felling of kanuka.”

“In one study, it was reported by Ramos and del Amo (1992) that consistent growth improvement and highest survival of all planted seedlings in a canopy treatment with 37% light transmission.”

Overuse of the passive voice

“We found greatest variability in growth and lower seedling survival from the forest edge sites.”

“Artificial gaps from selective felling of kanuka provided highest light transmission.”

“In one study, Ramos and del Amo (1992) reported consistent growth improvement and highest survival of all planted seedlings in a canopy treatment with 37% light transmission.”

What's wrong with these examples?

“Gap creation is likely to be, both through providing ideal sites for the growth of light demanding species like tōtara and through natural establishment of other future canopy trees, an important tool for restoring late successional canopy species in regenerating forests.”

“Similar to the effect of herbivory on tōtara seedlings planted into Pinus canopy gaps, survival in the edge sites, because of enhanced herbivory, most likely feral red deer which impacted some plants, was lower.”

Awkward phrasing

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“Similar to the effect of herbivory on tōtara seedlings planted into *Pinus* canopy gaps, survival in the edge sites, because of enhanced herbivory, most likely feral red deer which impacted some plants, was lower.”

Awkward phrasing

“Gap creation is likely to be an important tool for restoring late-successional canopy species in regenerating forests, both through providing ideal sites for the growth of light-demanding species like tōtara and through natural establishment of other future canopy trees.”

“Survival in the edge sites was lower because of enhanced herbivory, most likely feral red deer, which impacted some plants, similar to the effect of herbivory on tōtara seedlings planted into *Pinus* canopy gaps.”

What's wrong with these examples?

“We observed higher growth rates of tōtara seedlings under canopy gaps compared with the control, ringbarked, or forest edge treatments. This remained significant after 4 and 6 years.”

“Arrested successions typically occur when the growth of one species excludes another because they compete for the same resources. When this happens, restoration managers may need to actively intervene to help direct the succession toward the desired goal. An example of this is where low-light environments in early-successional stands prevent establishment of late-successional species (Li & Ma 2003).”

Unclear pronoun antecedent: “this”

“We observed higher growth rates of tōtara seedlings under canopy gaps compared with the control, ringbarked, or forest edge treatments. This remained significant after 4 and 6 years.”

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“We observed higher growth rates of tōtara seedlings under canopy gaps compared with the control, ringbarked, or forest edge treatments. This higher growth rate of tōtara seedlings remained significant after 4 and 6 years.”

“Arrested successions typically occur when the growth of one species excludes another because they compete for the same resources. When this happens, restoration managers may need to actively intervene to help direct the succession toward the desired goal. An example of this competitive exclusion is where low-light environments in early-successional stands prevent establishment of late-successional species (Li & Ma 2003).”

What's wrong with this example?

“If you want old farmland to turn back into a forest, then the best way is to cut down a bunch of trees that are there now to let more light in so that some other trees have a chance to grow tall. If you just plant new trees at the forest edge, or if you just slowly kill some of the tall trees by scraping off their bark, it won't work as good. You have to punch a hole in the forest that is there now.”

Too colloquial, non-academic tone

“If you want old farmland to turn back into a forest, then the best way is to cut down a bunch of trees that are there now to let more light in so that some other trees have a chance to grow tall. If you just plant new trees at the forest edge, or if you just slowly kill some of the tall trees by scraping off their bark, it won't work as good. You have to punch a hole in the forest that is there now.”

Too colloquial, non-academic tone

“Artificial gap-creation is the most effective means of restoring abandoned farmland to forest. These gaps provide the ideal sites for the growth of light-demanding species, which are otherwise excluded by the existing species. Other methods of forest-restoration, such as edge-planting and ring-barking, do not allow sufficient light transmission for the growth of these shade-intolerant species.”

Presentations can be more creative

- Scientific articles in journals usually follow a rigid formula
 - Errors are much clearer to readers
- However, presentations are much more diverse, and mistakes are more forgivable
 - You are presenting your research multiple times to different audiences
 - You have different amounts of time (e.g., 30-second, 3-minute, 15-minute talks, with time for questions; or, some listeners will stay at your poster longer than others)
 - The audience has a wide range of background understanding (e.g., some are experts, others are industry/government professionals, some are students)

Presenting a complete, consistent, coherent story



- Write a strong but specific title
 - Can be descriptive or declarative
- There may not be a need to go deep into the methods
- Draw focus on important aspects in the visualizations
- Remind us of your hypotheses and predictions when showing results
- Conclude with big picture implications
- Prepare for questions and interruptions

Let's look at some poster examples

- What do you like?
- What would you improve/change?
- A poster is half the story – there is also an oral presentation that accompanies it

Volume, morphology, and development of ice wedges in the Barrens of the Hudson Bay Lowlands, northern Manitoba

Tabatha Rahman*, Pascale Roy-Léveillé



INTRODUCTION

The Hudson Bay Lowlands (HBL) host Canada's southernmost continuous permafrost, and satellite imagery suggest that the 'Barrens' of the HBL host extensive ice-wedge polygon networks. Ice-rich permafrost terrain is particularly susceptible to **thaw-induced geomorphic, hydrological, and ecological change**. However, ground-ice research in the HBL remain scarce, **reducing the ability of communities** to mitigate risks associated with warming.

Research focus: Ice-wedge dynamics in the Barrens of the HBL.

Objective 1- Assess the volume, morphology, and activity of ice wedges;

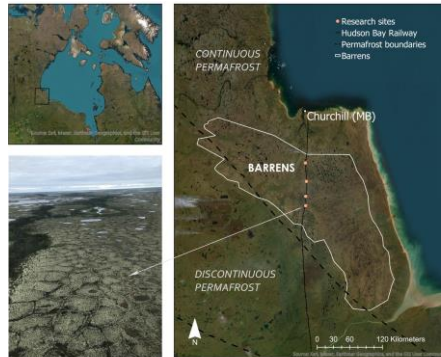
Objective 2- Reconstruct environmental conditions associated with ice-wedge development.

BARRENS OF THE HUDSON BAY LOWLANDS (HBL), MANITOBA

The Barrens are within the HBL, one of the world's largest permafrost peatlands.

They emerged ~6000 years ago from the Tyrrell Sea, due to isostatic rebound.

The area represents ~15 000 km² of lichen-covered peat plateaus with extensive ice-wedge polygon networks and thermokarst lakes and ponds.



METHODOLOGY AND DATA

Ice-wedge volume, morphology, and tridimensional distribution

- Drone and satellite imagery to map distribution
- Comparison of ice wedges seen on ground-penetrating radar (GPR; 2 & 4) and on aerial photographs for upscaling
- Permafrost coring (1 & 6) across ice-wedge troughs to estimate volume and 3-D distribution

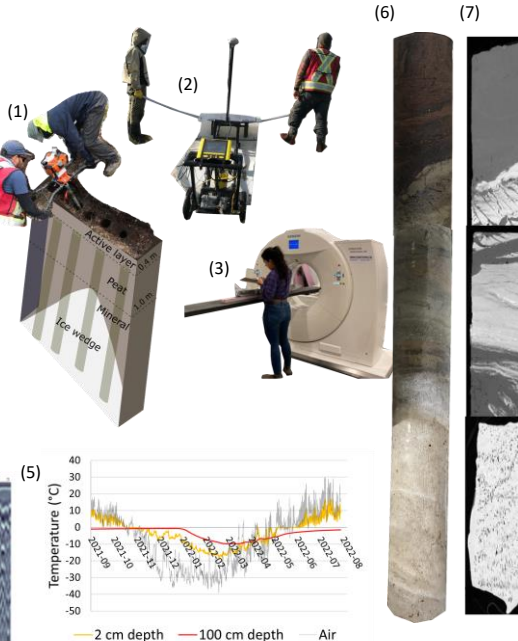
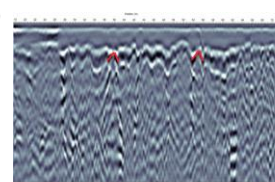
Ice-wedge activity

- Break cables and visual observations for signs of recent cracking
- Comparison of measured air and soil temperatures (5) to reported cracking thresholds

Environmental conditions associated with ice-wedge development

- Basal peat radiocarbon dating (¹⁴C) to determine time of emergence and peat development
- Analyses of cryostratigraphy via CT-scan imagery (3 & 7), stratigraphy via GPR (2 & 4) and core observations (6), soil macrofossils & pore ice chemistry, and ¹⁴C_{DOM} of wedge ice to determine timing of permafrost aggradation and wedge development
- Stable water isotope ($\delta^{18}\text{O}$ and D) analysis to elucidate the source of wedge-ice water.

(4)



PRELIMINARY RESULTS: Ice wedges in the Barrens...

Form widespread polygon networks that are well developed at the surface.



Ice-wedges may have been closer to the surface in the past. If so, when?

Are DEEP in permafrost! Over 1 m deep, while the active layer is < 40 cm.



Thaw of the permafrost surface will not likely melt wedge ice in the near-future.

Are not in peat! Top is either at or below the peat-mineral interface.



Did environmental or climatic conditions during peat aggradation halt ice-wedge growth?

Are about 1 m tall and 1.55 m wide.



This will be used to calculate wedge ice volume and potential landscape response to thaw.

Have a convex top! They are shaped like syngenetic wedges.



Is this the original shape, which we can use to infer growth processes, or has the shape undergone transformation?

ACKNOWLEDGEMENTS

This research is being conducted on Treaty 5 territory — the traditional land of the Cree, Dene, Inuit and Metis Nations. Thank you to Emmanuel L'Hérault, Adam Kirkwood, Frederic Brieger, Alex Janson, Sarah Gauthier, Samuel Gagnon, and Arianne B. St Amour. Thank you to our collaborators and funders:



SIGNIFICANCE

Results of **ice-wedge volume and tridimensional distribution** will allow for a better assessment of **terrain vulnerability** to thaw near crucial infrastructure in the Hudson Bay Lowlands, such as the Hudson Bay Railway, while supporting proactive adaptation to risks associated with rapid environmental change.

A better understanding of **the environmental conditions that led to ice-wedge development** and changes in ice-wedge activity in the Barrens will contribute to improving conceptual models of ice-wedge development in this region and in areas with similar conditions.

Water chemistry conditions are the strongest environmental predictors of fish communities in urban stormwater ponds

Gloria Wang, Donald Jackson

Department of Ecology & Evolutionary Biology, University of Toronto

Introduction

Stormwater ponds (SWPs) are artificial ponds widely used in newly developed areas to mitigate flooding and water pollution in urban streams by collecting runoff and retaining it between storm events^{1,2}

SWPs can be some of the only aquatic habitats in developed areas where wetland habitats have been lost due to urbanization³.

SWPs are not designed to be colonized by fish or support fish populations. Despite this, several fish species have been found using SWPs. With little research done on SWPs in Ontario, there is a gap in knowledge regarding how the water quality, design, and landscape characteristics of SWPs may affect fish biodiversity⁴.

Objective

Determine the relationship between environmental characteristics of SWPs and:

- 1) Fish species richness
- 2) Fish community compositions



Figure 1: Examples of stormwater ponds sampled (A, D), both minnow and crayfish traps used (B), and a pumpkinseed sunfish (C). Photo credit: Gloria Wang, Don Jackson

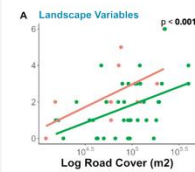
Methodology

- 50 SWPs in Brampton, Ontario were sampled from July to September 2022
- 4-5 sites sampled from each pond using **minnow traps** and **crayfish traps** (Fig. 1B)
- Fish were collected, identified, and counted 16-20 hours later

Variables

- **Landscape:** geographic information system (GIS) used to quantify land use zones in SWP catchment areas
- **Pond design:** GIS, field notes
- **Water quality:** vertical profiles and water samples taken July to September 2022
- **Negative binomial generalized linear models (GLMs)** modelled species richness with:
 - Landscape variables
 - Pond design variables
 - Water chemistry variables
- Best fit model assessed by **Akaike information criterion** for small sample size (AICc)
- **Canonical correspondence analysis** used to determine how environmental variables explain fish species abundance

Results



Negative Binomial GLM	df	AICc Score
Landscape	5	175.17
Pond design	6	178.22
Water chemistry	12	170.35

Table 1. AICc scores for the landscape, pond design, and water chemistry GLMs of fish species richness. Lower AICc score indicates better fit to the richness data.

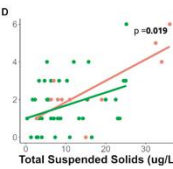
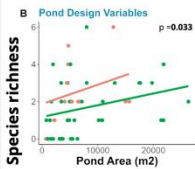


Figure 2. Linear regression of significant environmental variables in negative binomial GLMs at the landscape (A), pond design (B), and water chemistry (C, D) levels (n=50).

- 37 of 50 SWPs contained fish
- 10 fish species found (including invasive goldfish)

Significant predictors of species richness:

Road cover area positively associated with species richness (Fig. 2A)

Pond area positively associated with species richness (Fig. 2B)

Presence of submerged vegetation negatively associated with species richness (p = 0.042)

Nitrate negatively associated with species richness (Fig. 2C)

Total suspended solids positively associated with species richness (Fig. 2D)

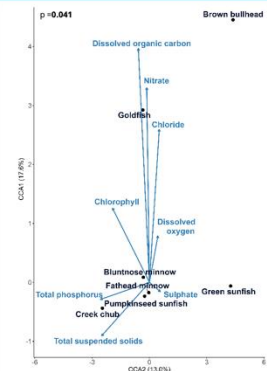


Figure 3. Canonical correspondence analysis of water chemistry variables associated with fish species abundance data displayed on 2 axes (n=37). Rare species (only found in 1 SWP) were excluded from analysis.

Significant predictors of community composition:

- Chloride**
p = 0.021
High levels: Brown bullhead, Goldfish
Low levels: Creek chub, Pumpkinseed sunfish
- Total suspended solids**
p = 0.038
High levels: Creek chub, Pumpkinseed sunfish
Low levels: Brown bullhead, Goldfish

Acknowledgements

Thank you to my supervisor, Dr. Don Jackson, and the members of the Jackson Lab for your support, advice, expertise, and insightful Friday Lab meetings, and for collecting much of the data for this project. This could not have been possible without you.

Thank you to Dr. Shelby Riskin and my classmates in EE848 for your encouragement, camaraderie, and thoughtful advice.

Thank you to the City of Brampton, Toronto and Region Conservation Authority, and the Credit Valley Conservation Authority for providing much of the environmental data and associated mapping files used in this project.

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Fish illustrations:
Source: B. (2020). National Conservation Training Center Publication.
Retrieved from US Fish & Wildlife Service. Licensed under CC0.

Poster created in Canva

Conclusions

- **Water chemistry conditions** best predicted species richness
- **Road cover, pond area, nitrate, & total suspended solids** were significantly associated with species richness:
 - These factors may be considered in SWP design and management to promote fish species richness and urban biodiversity
- **Chloride & total suspended solids** significantly predicted community composition
- **Chloride**, from road salt, was associated with community composition but not species richness, suggesting that other factors are greater drivers of richness (especially in the summer, when chloride levels are lowest)
- SWP outflow can potentially affect fish species in adjacent streams and lakes. Further research will inform best practices to mitigate impacts on downstream ecosystems⁵



Assessing the effectiveness of an R manual as a tool for undergraduate student learning

Deniz Raster, Dr. Shelby Riskin

Department of Ecology and Evolutionary Biology, University of Toronto

Introduction

Scholars are urging for an increase in quantitative skills and literacy among students in the disciplines of the biological sciences early on in their academic journey^{1, 2}.

R is an increasingly popular programming software commonly used for statistical analysis of data³. The growing popularity of R among researchers in ecology and evolution should be reflected in an increase in undergraduate-level courses teaching quantitative literacy using R^{1, 2}.

Accounting for the initial learning curve with R¹ and the limited availability of resources for beginners^{1, 2}, there is a need for a student-focused R resource². In this study, we implemented an R manual specifically designed for undergraduate students at the Ecology and Evolutionary Biology (EEB) department within the University of Toronto. The R manual was used as a learning resource in an undergraduate EEB course to help students navigate the programming software during laboratory units.

We hope to support and improve quantitative literacy among undergraduate students in EEB to ensure that they are well-equipped with the skills needed to become independent researchers.

Objectives

- 1) To assess if the R manual improves quantitative literacy among students
- 2) To assess if the R manual offers benefits as a teaching resource for potential users in the department and faculty

Materials and Methods

The R Manual

- The R manual was published online in the winter semester of 2023 on the EEB department's website

Survey Design

- We used student questionnaires at two time-points in the semester (pre-course and mid-course) to quantitatively assess student **experience**, **confidence**, **skills**, and **familiarity** in R, and assess student perceptions of the R manual in terms of skill **improvement**, **usefulness**, and **navigation**
- Graduate student Teaching Assistants (TA) and faculty were presented a Likert scale questionnaire to evaluate the perceived effectiveness of the R manual as a learning and teaching resource
- All questionnaires were distributed online through Qualtrics

Likert Scale Questions	Measure
Q4. I can import data into R.	R - Skills
Q5. I can make scatterplots using R.	R - Skills
Q6. I can calculate summary statistics of data using R.	R - Skills
Q12. The R manual improved my understanding of R.	Manual - Improvement

Figure 1. Portion of the Likert scale questions asked to students in both questionnaires (Q4, Q5, Q6) and in the mid-course questionnaire (Q12).

Statistical Analyses

- Kruskal-Wallis (K-W) tests were used to identify any significant differences in means with regards to students' R literacy pre- and mid-course
- Mann-Whitney U/Wilcoxon Rank-Sum tests identified any significant variables responsible for improved R literacy among students
- Dunn's Multiple Comparison tests with Bonferroni adjustments following K-W tests were used to identify any significant groups within significant variables

Discussion

- Differences in the means for questions 4, 5, and 6 only (measures of **skills**) in the pre- and mid-course questionnaires were statistically significant, indicating that students showed the most improvement in quantitative skills

- Prior to the lab component of the course, there was a significant difference in R **skills** between students who reported having experience versus those without. At the mid-course point, there was no significant increase for students who reported having prior experience. Students with no prior experience significantly increased their skills scores, reducing the gap between their scores and those with prior experience with R

- Students who reported that the R manual provided **improvement** to their understanding of R showed a significant increase in their **skills** scores

- Student perception of the R manual was generally positive, with the manual showing benefits in all categories assessed. The perception of TA and faculty of the R manual as a learning and teaching resource was positive, where faculty answered ('agree') nearly 100% of the time, while TA answered ('agree') > 72% of the time

- Based on the results of this study, we recommend that undergraduate students are introduced to R early in their academic journey. We also encourage the implementation of department- or faculty-specific R resources as they offer benefits to learning and teaching of quantitative skills using R

- This study was limited by the small sample size of the survey. Future studies would benefit from a larger sample size to fully capture the range of variables and mean scores

Results

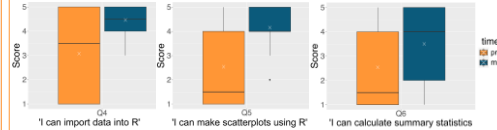


Figure 2. Likert scale scores for Q4, Q5, and Q6 (measure of R skills). Q4 $p < 0.01$; Q5 $p < 0.001$; Q6 $p < 0.05$; $n = 26$.

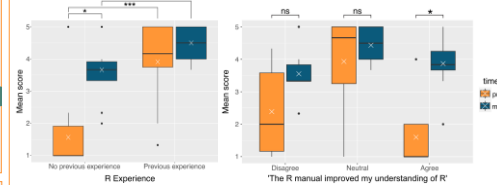


Figure 3. Correlation between mean scores for Q4, Q5, and Q6 and student previous experience with R. Square brackets indicate significant K-W tests. ** indicates $p < 0.01$, * indicates $p < 0.05$, no indicates no significance; $n = 26$.

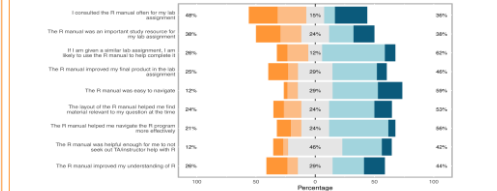


Figure 4. Correlation between mean score for Q4, Q5, and Q6 and agreement with Q12. Square bracket indicates significant K-W test, indicates $p < 0.05$; $n = 26$.

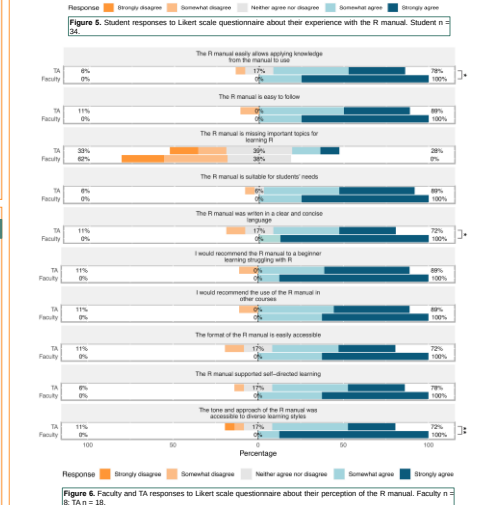


Figure 6. Student responses to Likert scale questionnaire about their experience with the R manual. Student $n = 26$.

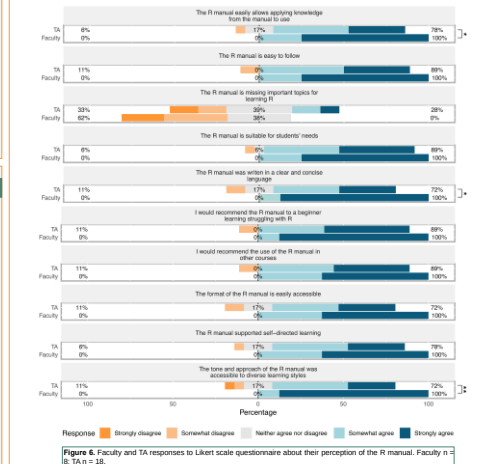


Figure 6. Faculty and TA responses to Likert scale questionnaire about their perception of the R manual. Faculty $n = 8$; TA $n = 18$.

Acknowledgements and Citations

I would like to thank my supervisor Dr. Shelby Riskin immensely for her invaluable support and guidance, Hana Fatahi for her time and support, and David Bost, for his graphic expertise. I also would like to thank the students, TA and faculty participants in the questionnaires, who volunteered their time to benefit learning at the University of Toronto.

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Does climate influence scale size among populations of *Anolis* lizards?

Katie Monat, Gavia Lertzman-Lepofsky, D. Luke Mahler

Background

- The scales of reptiles have been hypothesized to play roles in protection, thermoregulation, and hydoregulation^{1,2}.
- Previous studies exploring the effect of climate on scale size have found conflicting results, possibly because they did not consider the interaction between precipitation and temperature. The mechanisms driving scale morphology therefore remain unclear.
- To date, most studies of within-species patterns of scale size evolution have been limited in geographic and climatic scope.

Question

- How do environmental variables affect intraspecific patterns of scale morphology?

Methods

Study System

- Populations of four species of *Anolis* lizard across environmental gradients in the Dominican Republic (*A. cybotes*, *A. semilineatus*) and Puerto Rico (*A. cristatellus*, *A. stratulus*) (Figure 1).

Data

- Imaged dorsal skin of ethanol-preserved museum specimens.
- Measured size-corrected dorsal scale size using ImageJ.
- Conducted a principal component analysis (PCA) with climatic data from Worldclim 2.0⁴.
 - PC1 = temperature (explains 41% of variance) from cold to hot
 - PC2 = precipitation (explains 34% of variance) from dry to wet

Statistical Analyses

- Used mixed models to model the effect of environmental variables on mean scale area. We began by evaluating the following model, and used stepwise removal of higher-order terms (those without significant Satterthwaite *p*-values) to arrive at a final model:

$$\text{ln}(\text{mean.area}) \sim \text{PC1} + \text{PC2} + \text{PC1}^2 + \text{PC1}:\text{PC2} + (1|\text{locality})$$

Hypotheses

- Heat Dissipation Hypothesis:** because heat dissipates from the surface of scales³, and because larger scales have relatively greater surface area, large scales can act as a heat sink.

Prediction: populations of thermoregulating lizards in warmer environments should have larger dorsal scales to aid in thermoregulation.

- Interstitial Skin Hypothesis:** if more moisture is lost through interstitial skin (i.e., the skin between scales) than through scale surfaces, the evolution of larger scales (which reduce interstitial skin area) may mitigate moisture loss.

Prediction: populations in drier environments should have larger scales.



Figure 1. Map showing the location of the sampling sites and species on each island. Sites and species (*A. cybotes* & *A. semilineatus*) of the Dominican Republic are coloured blue. Puerto Rican sites and species (*A. cristatellus* & *A. stratulus*) are shown in red.

Results

- The data are broadly consistent with the heat dissipation hypothesis, although the effect of temperature on scale area depends on moisture in some species. In *A. cristatellus*, populations in warmer climates have larger scales (Figure 2C). A similar pattern is observed in *A. cybotes* and *A. stratulus* that occupy dry environments (Figure 2A, D).
 - In *A. cybotes* and *A. stratulus*, the interaction between temperature and precipitation modifies the effect of climate on scale size.
- The direction of the relationship between scale size and temperature may vary as a function of elevation in *A. cybotes* (Figure 2A).
- No significant effect of climate was detected on scale size in *A. semilineatus*, the sole species in our study that does not thermoregulate.

Species	Significant Model Terms
<i>A. cybotes</i>	PC1, PC1 ² , PC1:PC2
<i>A. semilineatus</i>	n/a
<i>A. cristatellus</i>	PC1
<i>A. stratulus</i>	PC1, PC1 ² , PC1:PC2

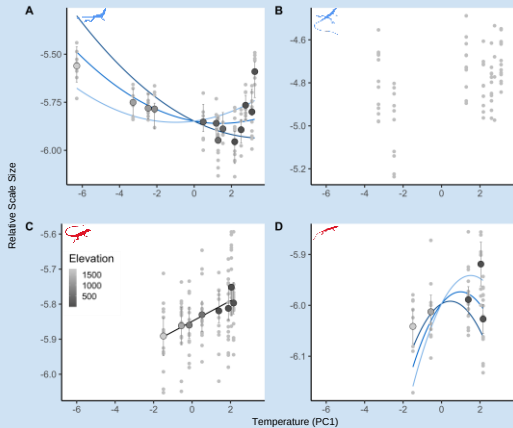


Figure 2. Plots showing the relationship between each PC axis and scale size for males of each species. Plots A to D show the relationship between PC1 and relative scale size from the model for each species: *A. cybotes* (A), *A. semilineatus* (B), *A. cristatellus* (C), and *A. stratulus* (D). Points represent predicted mean scale size from the model, and are coloured by elevation with 95% confidence intervals indicated by grey bars. Significant relationships between PC1 and scale size from the model are shown in blue. The interaction between PC1 and PC2 is represented by the blue lines, where dark blue reflects high precipitation and light blue reflects low precipitation.

Discussion

- We found evidence that scale size may evolve as a thermoregulatory structure in three anole species, as predicted if dorsal scales facilitate heat dissipation. We found little direct support for the interstitial skin hypothesis, but did find evidence that moisture interacts with temperature to shape scale evolution in two species (*A. cybotes*, *A. stratulus*).
- The role that scales play in thermoregulation may change across elevational gradients in *A. cybotes* (Figure 3).
 - At lower elevations, *A. cybotes* thermoregulate less³, and scales may act primarily to facilitate heat dissipation (to lower body temperature). Lowland *A. cybotes* populations have larger scales in warmer climates, which may help to prevent overheating.
 - At higher elevations, *A. cybotes* thermoregulate more by basking². During basking, large scales may facilitate heat gain³. Highland *A. cybotes* populations have larger scales in cooler climates, which may facilitate thermoregulation.
 - A. cybotes* is likely the only species that demonstrates this pattern because it is the only species we studied that lives at very high elevations.



Figure 3. Diagram showing the expected relationship between temperature and scale size across an elevation gradient.

Acknowledgements

I would like to thank all the members of the Mahler Lab for their help and support throughout the project, especially those that helped image the specimens that were used for this research: Martina Givov, Genevieve Kleinfisch, Brenda Li, and Michelle Su.

Citations

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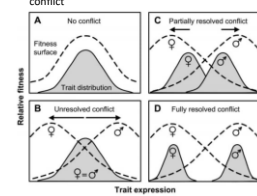
The effect of sex-limited evolution on the fitness of female *Drosophila melanogaster*

Youngseo Jeong, Aneil Agrawal

Introduction

Background

- Males and females share the autosomal genome, but they often have different fitness optima
- Sexually antagonistic selection arising from the shared genome but different fitness optima is like a tug-of-war
 - This form of sexual conflict is called intralocus sexual conflict



This experiment

- There is evidence for unresolved intralocus sexual conflict in populations of *Drosophila melanogaster*²
- We segregated chromosome III like a new sex-determining chromosome
 - No recombination and selection in only one sex

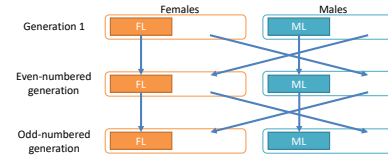
Predictions

- Sex limited evolution is expected to let the sexes evolve toward their respective fitness optima
- If the existing genetic variation is mostly sexually concordant and the sexual conflict is already resolved, then we expect to see the fitness of both males and females increase as a response to sex-limited evolution
- If the existing genetic variation is mostly sexually antagonistic, then we expect to see the fitness of females that carry the male-limited chromosome decrease, while the fitness of females that carry the female limited chromosome increase

Methods

Experimental evolution

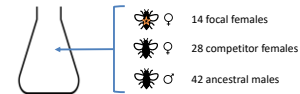
- 3 experimental populations and 2 control populations
- Chromosome III was evolved in a sex-limited way without recombination
- Evolved the populations for 9 generations using the selection scheme outlined below:



- Focal (FL) chromosomes evolved only in females and the Focal (ML) chromosomes evolved only in males
- Used two different TM3 balancers (TM3 Au and TM3 Ser) to prevent recombination and differentiate between the focal chromosomes
- In the control population, the FL and ML chromosomes spent equal amounts of time in both sexes, as well as with the two balancer chromosomes

Fitness Assay

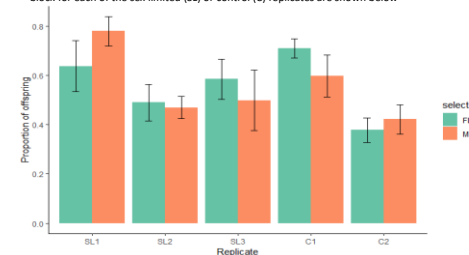
- Assayed the fitness of females carrying FL and ML chromosomes after 9 generations
- Estimated fitness by competing focal females with reference females, and measuring the ratio of offspring that were sired by the focal female
- All focal females carried either ML or FL chromosomes from the SL or C populations and one copy of TM3 Sb balancer chromosome.



- Ran a general linear model to the proportion of offspring produced by focal females to detect the effect of sex-limited evolution

Results

- The proportion of offspring produced by females with ML or FL chromosomes in the morning block for each of the sex limited (SL) or control (C) replicates are shown below



- The error bars represent standard error
- There was no significant selection treatment interaction effect ($p = 0.654$), but there were significant replicate and block effects
 - The afternoon block similarly showed no significant selection treatment interaction effect ($p = 0.644$)

Conclusion

- Did not find a significant relationship between sex-limited evolution and fitness, but this does not mean the effect of evolution is not present
- The uncertainty around fitness measures make it difficult to detect small fitness differences
- 9 generations of evolution is possibly too short to detect a change in fitness
 - Other experimental evolution experiments let populations evolve for at least 20 generations³
 - More generations and a more sensitive fitness assay could help detect the effect
- It is also possible that selection to adapt to the balancer genetic environment was strong for both males and females, making the direction of selection more sexually concordant
 - The ancestral population did not have balancer chromosomes
- When the population is moved away from its male and female fitness optima, sexual selection reinforces natural selection⁴

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Acknowledgements

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Synchrony between spring phenologies of vegetation and bird migration in an urban park

Jolie V.L. Nguyen¹, Henry James^{1,2}, Jonathan Ruppert^{1,2}, Marie-Josée Fortin¹

¹Department of Ecology and Evolutionary Biology, University of Toronto, ²Toronto and Region Conservation Authority



Background

- Migratory birds depend on stopover sites to replenish energy, rest, avoid adverse weather, and evade predation.¹
- During spring it is critical that migration happens in synchrony with vegetation green-up, alongside insect emergence, at these stopover sites.²
- As the spring phenology of bird migration and green-up may be changing due to climate change,¹ there is a possibility for phenological mismatches.²

Objectives:

- Investigate how the spring phenology trends of bird migration and vegetation green-up are changing over time
 - Compare the phenological synchrony between spring vegetation green-up and bird arrival
 - Investigate the difference in phenological synchrony between males and females of the same species

Methods

Study Area

- Tommy Thompson Park (TTP) - an urban stopover site in Toronto + Two wider regional scales: coastal lake area and part of the Greater Toronto Area (GTA)



Fig 1. Map of study areas: GTA, coastal, and TTP, with a close-up of TTP. The star indicates the location of the Bird Research Station.

Bird Banding Data

- Data collected by the Toronto and Region Conservation Authority (TRCA) during spring migration (April-June) at TTP from 2004-2021



Fig 2. A Myrtle Warbler caught for banding at TTP

Table 1. The 5 Focal species; 3 warblers and 2 kinglets

Common Name	Scientific Name	Abbreviation	n
Black-throated blue warbler	<i>Setophaga caerulescens</i>	BTBW	284
Magnolia warbler	<i>Setophaga magnolia</i>	MAWA	1568
Myrtle (yellow rumped) warbler	<i>Setophaga coronata coronata</i>	MYWA	3387
Ruby-crowned kinglet	<i>Corthylio caeruleus</i>	RCKI	1299
Golden-crowned kinglet	<i>Regulus satrapa</i>	GCKI	1617

Vegetation Phenology Data

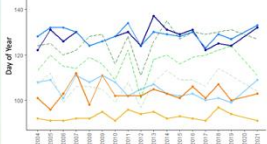
- Mean green-up day in each area for 2004-2021 was calculated using Moderate Resolution Imaging Spectroradiometer (MODIS) data at 500m resolution

Analysis of Phenology and Synchrony

- Trend analysis of green-up day, first bird arrival day, median bird arrival day, differences between male and female arrival
- Comparison between green-up day and bird arrival, calculating how the difference in timing between the two has changed over time

Results

First Arrival Day

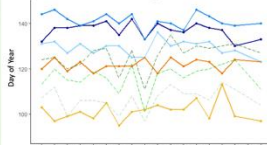


The mean green-up day did not significantly change to become earlier or later but was subject to year-to-year fluctuations.

Only the first arrival day of Myrtle Warblers (MYWA) significantly changed, arriving to TTP earlier in spring.

The median arrival day of each of the five focal species did not change significantly.

Median Arrival Day



SPESIES GREEN-UP DAY

Warblers
BTBW
MAWA
MYWA
Kinglets
RCKI
GCKI

— TTP
— Coast
— GTA

Fig 3. First arrival and median arrival by species at TTP from 2004-2021. The mean green-up day for TTP, the coastal area, and the GTA are also shown.

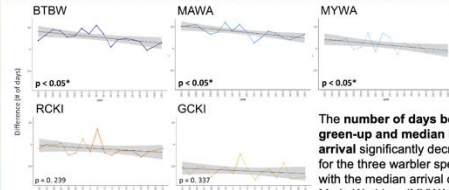


Fig 4. Linear regression analysis and plot of the number of days between mean green-up day and median bird arrival day.

The number of days between green-up and median bird arrival significantly decreased for the three warbler species, with the median arrival date of Myrtle Warblers (MYWA) now becoming earlier than green-up.

Males arrived earlier than females for the five species, typically closer to or before green-up.

The difference between the median arrival day of the male and female Ruby-crowned Kinglets (RCKI) significantly decreased from 2004-2021, with males arriving later and females arriving earlier.

Male vs. Female Arrival at TTP

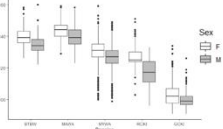


Fig 5. Distribution of the arrival of all males and females of the 5 species at TTP

Conclusions

Changing trends were found:

- The first arrival day for Myrtle Warblers became earlier.
- The number of days between green-up and median bird arrival decreased for the warbler species, with the birds arriving closer to or before green-up day.
- Males generally arrive earlier than females, which could make them more vulnerable to changing phenological trends.

The spring phenologies of vegetation and different migratory bird species may be shifting at different rates, potentially giving rise to asynchronous phenological responses.

Further studies should investigate how this asynchrony could negatively impact the success of bird migrations.

Acknowledgements

Thank you TRCA for the data. Thank you to Amanda Schwantes for help with the MODIS data. Thank you Marie-Josée Fortin and Jonathan Ruppert for being great supervisors, as well as Henry James for all the help. Finally, thanks to the entire EE8488 class for the support throughout.

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2. Kellermann J, van Riper C. 2012. Detecting mismatches of bird migration stopover and tree phenology in response to changing climate. *Decoecology*, 178, 1227-1238.



Do broader heads make for a stronger bite?

A biomechanical analysis of a putatively ecologically dimorphic trait in the northeastern pine sawyer beetle (Cerambycidae: *Monochamus notatus*)

Wei Han Lau¹, Rowan French¹, Alexander Tininus¹, Locke Rowe¹ | ¹ University of Toronto, EEB

Introduction and questions

- **Ecological dimorphisms** are sexual dimorphisms that are driven by natural selection rather than sexual selection¹.
- In *Monochamus notatus*, females have markedly **wider** heads than males.

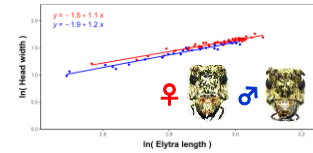


Fig. 1. Statistics for a two-tailed t-test for the difference between the estimated slope coefficients and a hypothetical slope value predicted from isometry: Head width scales hyperallometrically in both males (null hypothesis slope = 1, $P < 0.01$) and females (null hypothesis slope = 1, $P < 0.01$). Females have wider heads than males of the same size (head width/elytra length, male mean = 0.240 cm, female mean = 0.256 cm, $P < 0.01$).

Q: Is head width an **ecologically dimorphic** trait?

- Female *M. notatus* use their mandibles to **chew oviposition slits** in the bark of freshly fallen conifers¹.
- **Hypothesis**: Broader heads in female *M. notatus* could give rise to a stronger bite force in females.
- Other aspects of **internal** (musculature) and **external** (mandibles) anatomy could also contribute to a stronger bite force in females.

Q(a): Does head shape (and its associated traits) **give rise** to a **difference** in bite force between the sexes?

- Comparing how bite force and traits that contribute to bite force **scale** with **body size** can reveal whether bite force is influenced by a difference in traits or simply by a change in overall size.
 - **Isometry** – a 1:1 scaling relationship with body size.
 - **Hyperallometry** – When a trait has a higher than proportional relationship with body size ($> 1:1$).
 - **Hypoallometry** – When a trait has a lower than proportional relationship with body size ($< 1:1$).

- **Prediction**: Females have a **larger bite force** than males across the body size spectrum.

Q(b): If a difference in bite force exists between the sexes, how is that **difference** in bite force **achieved**?

- **Prediction**: Females have a **larger muscle volume** and/or a **larger mechanical advantage** and/or **smaller muscle pennation angles** and/or **shorter muscle fibre lengths** than males across the body size spectrum.

Methods

- **80** ethanol-preserved specimens from Janvier, Alberta.
- **Subset of 10 males and 10 females** chosen for micro-CT analysis of mandibular adductor muscles.

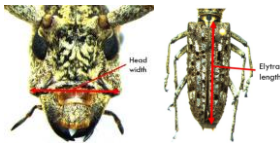


Fig. 2. Linear measurements of external (left: head, right: body) morphology

- **Linear measurements** of external morphology: elytral length, head width, mandibular in-lever and out-lever.
- **20 specimens soaked for 5 days in 2% phosphotungstic acid (PTA)** to improve tissue contrast in the Micro-CT scans⁴.
- **Micro-CT analysis**: muscle volume, muscle pennation angle, muscle cross-sectional area.

Results

At smaller sizes, females have a stronger bite force and larger muscle volumes than males. However, the opposite is true at larger sizes.

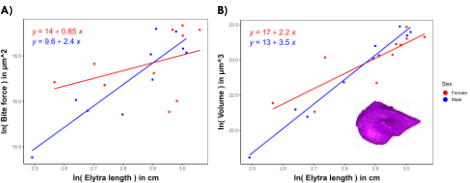


Fig. 3. Statistics for a two-tailed t-test for the difference between the estimated slope coefficients and hypothetical slope values predicted from isometry: A) Bite force scales hyperallometrically in males (null hypothesis slope = 2, $P < 0.01$) and hypoallometrically in females (null hypothesis slope = 2, $P < 0.01$). The slope of males and females differ from each other ($P < 0.01$). B) Muscle volume scales hyperallometrically in males (null hypothesis slope = 3, $P < 0.01$) and hypoallometrically in females (null hypothesis slope = 3, $P < 0.01$). The slope of males and females differ from each other ($P < 0.01$).

Mechanical advantage scales negatively in both sexes, but this pattern is stronger in females. Muscle fibre length does not contribute to the observed pattern in bite force between the sexes.

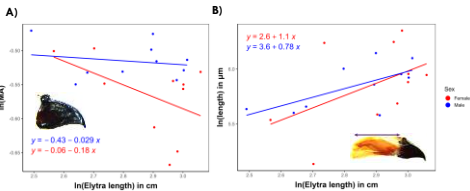


Fig. 4. Statistics for a two-tailed t-test for the difference between the estimated slope coefficients and hypothetical slope values predicted from isometry: A) Mechanical advantage scales hypoallometrically in both males (null hypothesis slope = 0, $P < 0.01$) and females (null hypothesis slope = 0, $P < 0.01$). The slope of males and females differ from each other ($P < 0.01$). B) Muscle fibre length scales hypoallometrically in males (null hypothesis slope = 1, $P < 0.01$) and isometrically in females (null hypothesis slope = 1, $P = 0.656$). The slope of males and females do not differ from each other ($P = 0.220$).

Conclusions

- Females have a **size-dependent** performance advantage in bite force over males. Head width could be an ecologically dimorphic trait that enables **smaller** females to accommodate a **higher** muscle volume.
- This pattern in bite force is **achieved** by: 1) a **smaller** than proportional investment in **muscle volume** and 2) a stronger **negative** relationship between **mechanical advantage** and body size in females.
- At small body sizes, females may be subject to stronger selection on bite force than males to meet functional demands related to oviposition slit chewing.



Literature cited



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But how do you actually make a poster?

- PowerPoint:
 - Design -> slide size -> change slide dimensions
- Canva:
 - Create a Design -> search for “poster” -> select size
 - Can be resized as well (Resize button, top left)
- Sizing tips:
 - Set zoom to 100% when editing
 - Recommend 36 x 24 inches
 - Make sure to add content can be scaled (e.g., zooming in won't reduce quality)

Recap

- Presentations should be accessible to a wide audience
- For reports, start broad in the introduction, narrow into your focus, and end broad in your conclusions
- Ensure that reports, including posters, are free of grammatical and scientific errors, and present the most complete story in a clear way